# Tutorial Problem Set XI

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*The topics of this tutorial were antiderivatives, Riemann sums, definite integrals and the fundamental theorem of calculus. Furthermore, Quiz 5 was written.*

**Problem 1.** Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Prove that

$$F(x) = \int_0^x f(t)\, dt$$

is an antiderivative of $f$.

**Problem 2.** Let us now consider a functions that is not Riemann integrable.

(a) Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{otherwise} \end{cases}$$

is *not* Riemann integrable on $[0, 1]$.

(b) Show, however, that it is still true that the following limits exist and are equal:

$$\lim_{n\to\infty} \sum_{k=1}^{n} f\left(\frac{k}{n}\right) \cdot \frac{1}{n} = \lim_{n\to\infty} \sum_{k=1}^{n} f\left(\frac{k-1}{n}\right) \cdot \frac{1}{n}.$$

Contemplate on why this does not contradict that $f$ is not integrable.

**Problem 3.** Prove that

$$\lim_{n\to\infty} \left( \sum_{k=1}^{n} \frac{1}{n+k} \right) = \ln(2).$$

**Problem 4.** Let $f : [0, 1] \to \mathbb{R}$ be a continuous and non-negative function such that

$$\int_0^1 f(x)\, dx = 0.$$

Prove that $f(x) = 0$ for all $x \in [0, 1]$.

**Problem 5.** Suppose that $f : [0, 1] \to \mathbb{R}$ is a differentiable function satisfying that $\int_0^1 f(x)\, dx = 0$. Prove that

$$\int_0^1 x f'(x)\, dx = f(1).$$